

Advancing our understanding of galaxy structure evolution: The critical role of gas in disrupting young clusters

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Abstract

It is believed that at least 50% of the stars in spiral galaxies like the Milky Way form within dense embedded clusters that are observed to cover a mass range of 10-106 solar masses. Observations and cosmological simulations agree that gradients in the local gas density play an important role in the early evolution of clusters at the high mass end of the cluster mass distribution. However, current cosmological zoom-in simulations of galaxy evolution are unable to resolve low mass clusters, making it impossible to verify if low mass clusters are subject to the same disruptive mechanism as high mass clusters. We propose to build N-body simulations of low mass clusters embedded within a realistic galactic potential that includes the contribution from the surrounding gas. The galactic potential will be interpolated from snapshots of cosmological simulations of spiral galaxies, which will allow us to compare the kinematic evolution of the low-mass clusters with the gas distribution from the cosmological simulation. The embedded cluster mass function is dominated by clusters at the low mass end, making the physics governing the lower mass limit critical to our understanding of stellar mass growth in spiral galaxies. The proposed hybrid simulations will enable a novel theoretical study that is needed to accurately interpret the complex physics of galaxy evolution using HST and JWST data of both resolved and unresolved spiral galaxies.

■ Scientific Justification

The dissolution of young stellar clusters in the FIRE simulations: measuring the impact of cold ISM.

The disc of the Milky Way (MW) is constantly being populated through the disruption and dissolution of stellar clusters with masses between 10 and $10^6 M_\odot$. Both recent (e.g., Quintana et al., 2025) and seminal (Lada and Lada, 2003) studies of clusters within our galaxy suggest that between 50% and 90% of the stars of the MW disc were formed within $10 - 10^2 M_\odot$ clusters that were disrupted rapidly after being formed. As spiral galaxies such as the MW are actively forming clusters from their dense cold gas, quantifying the processes through which low-mass stellar clusters disrupt is essential to understand how the complex structures of spiral galaxies like our own evolve.

However, we are not yet able to resolve low-mass ($< 10^3 M_\odot$) young clusters within observations of external galaxies (Thilker et al., 2025) nor within realistic simulations of galaxies (e.g., Grand et al., 2024; Hopkins et al., 2018). Additionally, young clusters are often embedded within dense, dusty and gaseous environment, making their observation even more difficult. Future cycles of JWST will allow to identify young embedded stellar population in external galaxies (Thilker et al., 2025) but will not be able to resolve low-mass clusters. To overcome this limitation, **it is imperative to provide the community with a simulation-based framework to quantify the disruption rates of young clusters** that our current theories predict for different mechanisms, so their observables can be constrained to ultimately better interpret future observations.

We propose to build N-body simulations of clusters with masses between 10 and $10^3 M_\odot$ embedded within a realistic galactic potential interpolated from a MW-like galaxy from the FIRE simulations (Hopkins et al., 2018) to quantify the role of gas in disrupting young clusters. This study will mark an important step to quantify the role of gas in disrupting young clusters as well as to constrain the origin of young stellar associations as remnants of disrupted clusters. By providing results that will interpret the structure of unresolved and resolved spiral galaxies within JWST images, **this proposal will extend the legacy of JWST archival data for one JWST main tasks: understand how galaxies form and evolve.**

Our understanding of cluster disruption is critically limited by mass resolution.

Since the seminal work of Lada and Lada (2003), important efforts have been made to understand observationally how clusters disrupt within the MW using the few dozens of embedded clusters for which infrared data is available (e.g., Kuhn et al., 2019). To overcome the observational biases related to MW clusters, Thilker et al. (2025) combined high resolution surveys such as HST and JWST to characterize clusters in external galaxies. However, although it is expected that future cycles of JWST will allow to detect young stellar population that is deeply embedded, JWST does not possess the resolution necessary to detect

low-mass ($< 10^3 M_\odot$) compact clusters. It is then critical to provide the community with a simulation-based framework to overcome the problem of resolution when investigating how young clusters disrupt through their interaction with their surrounding gas.

A recent effort to infer the role of gas in disrupting clusters was made by [Jørgensen and Church \(2025\)](#) who used a feedback-free galactic model and estimated that interactions with giant molecular clouds contribute 7 to 8% to the total mortality rate of young clusters. However, they did not include clusters with masses $< 600 M_\odot$. Similarly, although typical realistic simulations of MW-like galaxies such as FIRE ([Hopkins et al., 2018](#)) or AURIGA ([Grand et al., 2024](#)) include the important component of feedback from supernovae (SNe), they do not allow for the study of low-mass clusters as their mass resolution is $\sim 10^3 M_\odot$. It is then clear that to build a simulation-based framework to understand how young low-mass clusters are formed, N-body simulations of such clusters within a realistic simulated galaxy that includes gas and feedback are needed. This proposal will deliver such essential, yet missing, ingredient.

This proposal: simulate young low-mass clusters within a realistic galaxy to characterize their disruption.

We plan to characterize the disruption of young clusters by building N-body star-by-star simulations of low-mass ($10 - 10^3 M_\odot$) stellar clusters embedded in a galactic potential interpolated from snapshots of MW-like galaxies from the cosmological hydrodynamical zoom-in simulations of the FIRE collaboration ([Hopkins et al., 2018](#)). The realistic nature of the galaxies from FIRE make them perfect candidates to build a framework to interpret observables of future JWST cycles that will be critical to understand how young clusters evolve through interacting with their gaseous galactic environment. For this reason, this proposal is relevant to extend the legacy of JWST archival and future data to understand how galaxies evolve.

We propose to use simulated galaxies from FIRE to: **(i)** generate the initial conditions for young low-mass ($10 - 10^3 M_\odot$) clusters by sampling from the initial positions and velocities of high-mass ($10^4 - 10^6 M_\odot$) clusters previously identified within the FIRE galaxies; and **(ii)** model the local galactic potential by applying the code COGSWORTH ([Wagg et al., 2025](#)) to FIRE’s snapshots. We plan to use the code PETAR ([Wang et al., 2020](#)) to perform intra-cluster N-body star-by-star integration of the generated clusters. After integrating the clusters in time and measuring their virial ratio at each time-step, we plan to measure their distances to their closest SN feedback shock front at the moment of being disrupted to infer to which extent such steep gradient in the gas density could participate in disrupting such clusters. By doing so, we will be able to complete the low-mass end of Figure 1, which show the distribution of the distances to the shock front when being disrupted for different cluster masses.

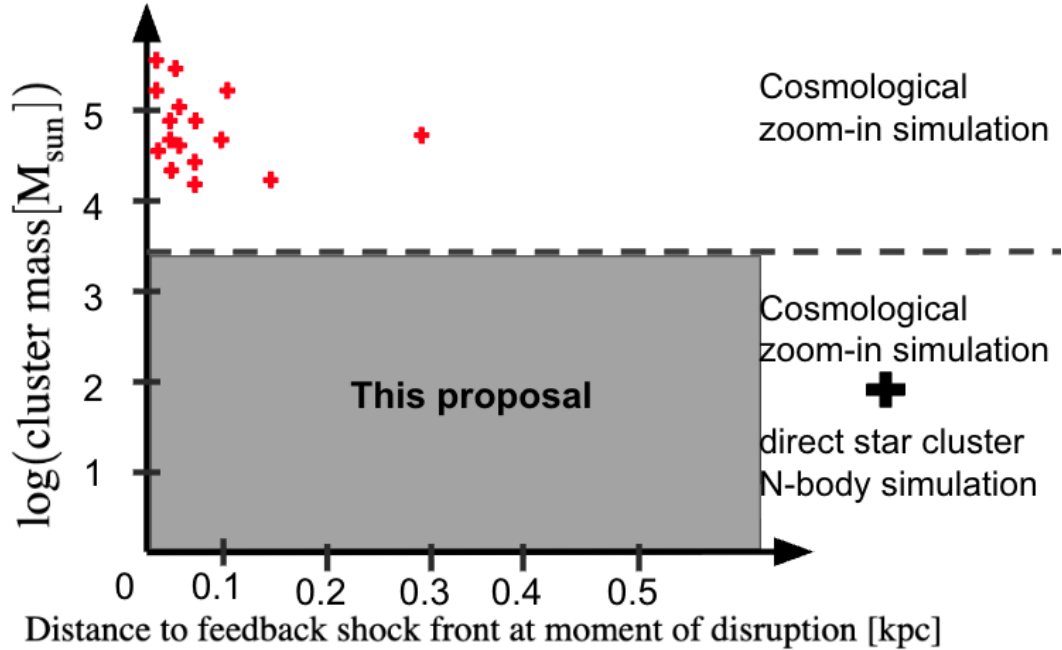


Figure 1: **Complex gas structures in a galaxy highly impacts the disruption of massive clusters and hence galaxy evolution.** The plot shows the masses of massive clusters identified within galaxy simulations from FIRE (Hopkins et al., 2018) as a function of their distance to their closest feedback shock front at the moment of disruption, quantifying the impact of gas in disrupting clusters. This proposal will complete the low-mass end of this plot.

The results of this proposal will quantify the role of gas in disrupting clusters in different mass ranges, which is critical to understand how disruption occur at different scales. Ultimately, this proposal will provide essential tools to solve the problem of resolution in the study of the disruption of young stellar clusters.

This proposal will advance our understanding of the evolution of the structure of spiral galaxies

This proposal will characterize how young stellar clusters are disrupted through their interactions with their surrounding gas, which is a missing yet critical piece of our modeling of cluster disruption. Such results are fundamental to understand how the continuous formation and disruption of clusters in spiral galaxies impacts the evolution of galactic structure.

The time is ripe for such study to be performed as it will allow us to build a framework that will provide tools to interpret data from future JWST images of deeply embedded extra-galactic young clusters, before such data is obtained. In this way, **this proposal will expand our understanding of how the life-cycle of stars impact the structure of their host galaxies and hence their evolution over time, which is one of the main objectives of JWST and HST.** This work will extend the legacy of both JWST archival and future data by providing tools useful to the community to better model the dynamics of young clusters in a galactic and ISM context.

References

- Robert J. J. Grand, Francesca Fragkoudi, Facundo A. Gómez, Adrian Jenkins, Federico Marinacci, Rüdiger Pakmor, and Volker Springel. Overview and public data release of the augmented Auriga Project: cosmological simulations of dwarf and Milky Way-mass galaxies. *Monthly Notices of the Royal Astronomical Society*, 532(2):1814–1831, August 2024. doi: 10.1093/mnras/stae1598.
- Philip F. Hopkins, Andrew Wetzel, Dušan Kereš, Claude-André Faucher-Giguère, Eliot Quataert, Michael Boylan-Kolchin, Norman Murray, Christopher C. Hayward, Shea Garrison-Kimmel, Cameron Hummels, Robert Feldmann, Paul Torrey, Xiangcheng Ma, Daniel Anglés-Alcázar, Kung-Yi Su, Matthew Orr, Denise Schmitz, Ivanna Escala, Robyn Sanderson, Michael Y. Grudić, Zachary Hafen, Ji-Hoon Kim, Alex Fitts, James S. Bullock, Coral Wheeler, T. K. Chan, Oliver D. Elbert, and Desika Narayanan. FIRE-2 simulations: physics versus numerics in galaxy formation. *Monthly Notices of the Royal Astronomical Society*, 480(1):800–863, October 2018. doi: 10.1093/mnras/sty1690.
- Timmi G. Jørgensen and Ross P. Church. The young stellar clusters in M51 and the impact of GMC encounters. *Monthly Notices of the Royal Astronomical Society*, 543(2):1410–1428, October 2025. doi: 10.1093/mnras/staf1548.
- Michael A. Kuhn, Lynne A. Hillenbrand, Alison Sills, Eric D. Feigelson, and Konstantin V. Getman. Kinematics in Young Star Clusters and Associations with Gaia DR2. *The Astrophysical Journal*, 870(1):32, January 2019. doi: 10.3847/1538-4357/aaef8c.
- Charles J. Lada and Elizabeth A. Lada. Embedded Clusters in Molecular Clouds. *ARA&A*, 41: 57–115, January 2003. doi: 10.1146/annurev.astro.41.011802.094844.
- Alexis L. Quintana, Emily L. Hunt, and Hanna Parul. How many stars form in compact clusters in the local Milky Way? *Astronomy & Astrophysics*, 701:L2, September 2025. doi: 10.1051/0004-6361/202556366.
- David A. Thilker, Janice C. Lee, Bradley C. Whitmore, Daniel Maschmann, Kiana Henny, Rupali Chandar, Daniel A. Dale, Sinan Deger, Médéric Boquien, Aida Wofford, Leonardo Úbeda, Alessandro Razza, Ashley T. Barnes, Francesco Belfiore, Frank Bigiel, Kathryn Grasha, Brent Groves, Hwihyun Kim, Ralf S. Klessen, Justus Neumann, Francesca Pinna, M. Jimena Rodríguez, Erik Rosolowsky, Eva Schinnerer, and Thomas G. Williams. PHANGS-HST Catalogs for $\sim 100,000$ Star Clusters and Compact Associations in 38 Galaxies. II. Physical Properties from Decision-tree-based Spectral Energy Distribution Fitting of NUV-U-B-V-I Photometry with Categorical Priors Set by $H\alpha$ Emission, Cluster Morphology, and Other Auxiliary Information. *The Astrophysical Journal Supplement Series*, 280(1):1, September 2025. doi: 10.3847/1538-4365/addabb.
- Tom Wagg, Katelyn Breivik, Mathieu Renzo, and Adrian M. Price-Whelan. cogsworth: A Gala of COSMIC Proportions Combining Binary Stellar Evolution and Galactic Dynamics. *The Astrophysical Journal Supplement Series*, 276(1):16, January 2025. doi: 10.3847/1538-4365/ad8b1f.

Long Wang, Masaki Iwasawa, Keigo Nitadori, and Junichiro Makino. PETAR: a high-performance N-body code for modelling massive collisional stellar systems. Monthly Notices of the Royal Astronomical Society, 497(1):536–555, September 2020. doi: 10.1093/mnras/staa1915.